TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Cableless electrocardiographic device for ICU or wearables**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical / Electronics		Bio / Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Measuring Instruments	Measurement Sensors and Components	Therapeutic Devices & Diagnostic Devices	Bio-signal Measurement / Diagnostic Devices
Concept of Proposed Technology	Conventional medical ECG (electrocardiogram) devices require a shielded cable to connect every electrode to the recorder or monitor. This approach is bulky for wearables, and, in ICU, results in messy setup and risk of cross contamination, since the cables are usually too expensive to be disposable. We propose to thoroughly revisit the ECG metrology, so that (active) electrodes could be connected in the simplest way as physically possible: with one single shared unshielded wire. In addition to simplifying the setup dramatically, such approach also opens the way to new formfactors and products with unique innovative features, such as, for example, the possibility to connect electrodes anywhere along a disposable wire, in a modular way for neat setups with disposable low-cost devices.			
Rationale for Joint R&D Project	Electronics can be efficiently integrated into low cost formfactors. Unfortunately, this advantage is not fully exploited in ECG devices where the cabling burden remains the same as when the electronics started more than 100 years with vacuum tubes. We have conceptualized novel approaches that address this issue. However, development is needed for demonstration and verification. To this aim we need a joint R&D project with analog IC designers and/or partners with marketed ECG medical devices or wearables.			
Key Words	1. ECG (electrocardiograph) 2. analog IC 3. medical device 4. wearables			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 2	End of the Project	Level 4
R&D Objectives	The primary objective of the project is to develop electronic technology for a 12-lead ECG device to have its electrodes connected to the same single unshielded wire so that demonstration and verification against the medical standards can be performed. Other objectives include the integration of this new technology in different form factors, such as, e.g., a wearable for long term recordings, and a defibrillation-proof ICU monitor with disposable applied part so that cross-contamination between patients is reduced to its minimum (today, 50% of the cleaned cables still carry pathogens according to a clinical study).			

R&D Target Specifications	The new ECG circuitry to develop shall:
	• be **modular**, from 1 to 12 leads
	• have active electrodes connected to a **single shared unshielded wire** (no lead cable)
	• be low cost, so that it can be competitive.
	  ○ as disposables in clinical use (e.g., ICU) thereby suppressing the need for ECG lead cleaning.
	  ○ to be easily integrated into washable garments
	• comply with medical ECG standards (in particular, noise, bandwidth, defibrillation proof, etc.):
	  ○ 60601-2-25 (diagnostic ECG)
	  ○ 60601-2-27 (ECG monitor)
	  ○ 60601-2-47 (ambulatory ECG, Holter)
	• include electrode impedance monitoring.
	• have a single power supply (e.g., USB)			

R&D Plan	Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives. WP1: Architecture (M0-M6) The architecture of the device will be detailed from the initial concept. Simulations at system level that includes the body in the loop as well as prototyping with discrete components will be performed so that dimensioning and performance verification can be done at system level. The system specification and component requirements for the active-electrode IC, recorder, textile, dry electrode, etc. will be the output of this WP. M1: Architecture and component requirements WP2: Design (M7-M12) The active-electrode IC will be designed as well as the other electronics. The textile will be selected and the design of the garment produced. The active-electrode packaging design will be detailed. M2: Detailed design WP3: Prototyping (M13-M18) The components will be manufactured and assembled according to the design. M3: Prototype			

	WP4: Tests (M19-M24) Tests will be performed to verify the new technology with respect to the medical standards.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	CSEM role and responsibility will be in the • architecture of the new ECG circuitry to be developed, • active-electrode IC requirements specification, • medical compliance of the device, • development of the wire controller at the recorder, • integration of the active-electrode IC, • system verification and tests
Expected R&R of Korean SME Partner	The expected roles and responsibilities of Korean SME will be either, or both, in the • development of the active-electrode IC and its verification, • packaging and integration of the active-electrodes in textiles resulting in low-cost washable garments

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will demonstrate a new disruptive approach measuring ECG, particularly advantageous for modular, low-cost, and disposable applied part, reducing the risks of nosocomial cross-contamination in clinical settings, and reducing the integration cost in washable wearables. From the demonstration, companies producing ICs, wearables, or medical devices will integrate this innovative technology and take large market share of the ECG business. In a second step, the same approach can be extended to other biopotentials. One with significant current and future market is the EEG (electroencephalogram). Thanks to the advantages of the proposed technology, new products, e.g., neater, with higher performance, lower cost—can be designed.

6. References

(1)

(2)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization